Cambodia Rice Yield Prediction

A Comparative Study of Machine Learning Models

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Introduction

Why Cambodia rice yield?

- Rice is Cambodia's staple crop and largest agricultural export
- Yield fluctuations directly impact food security and rural livelihoods
- Country-aggregated annual data is publicly available from FAOSTAT and the World Bank

Objectives

- Build a merged dataset from FAOSTAT + World Bank climate data
- Train and compare 3 ML models
- Evaluate under both chronological and cross-validation protocols

Tasks

Regression: Predict exact yield (kg/ha)

Classification: Predict High or Low yield (above/below median)

Dataset

3 sources merged on year (1990–2024):

- **FAOSTAT** — area harvested, production, yield
- **World Bank** — annual mean temperature (°C)
- **World Bank** — annual precipitation (mm)

Production was dropped: yield = production / area, so it would leak the target.

Feature	Unit
Area harvested	ha
Mean temperature	°C
Precipitation	mm
Target (reg.)	kg/ha
Target (cls.)	High / Low

Dataset size

35 rows · 3 features · 1 target | Chronological split: 28 train (1990–2017) / 7 test (2018–2024)

Classification threshold: median computed on **training set only** (no leakage)

Data overview

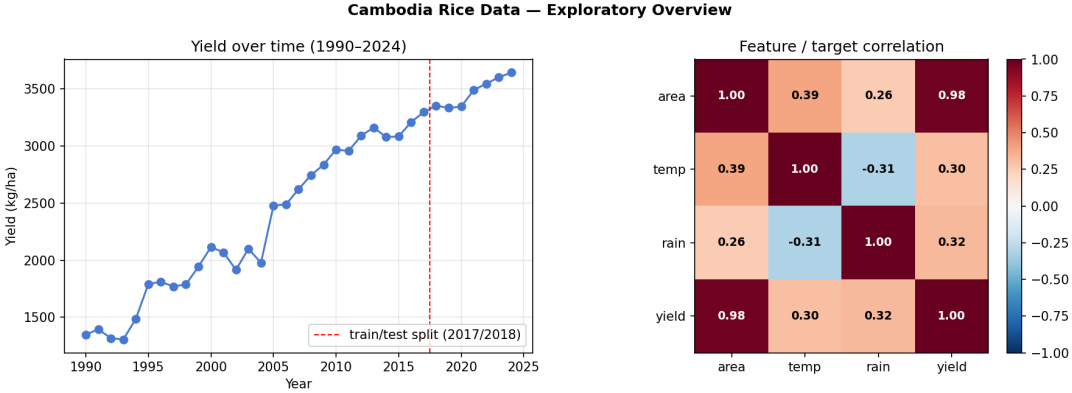


Figure: Yield grew strongly from 1990 to 2024; area-harvested has the highest correlation with yield ($r = 0.98$).

Method 1 — Linear Regression

- Predicts yield as a weighted sum of features:

$$\hat{y} = \mathbf{w}^\top \mathbf{x} + b$$

- Fitted via Ordinary Least Squares (sklearn)
- Simple, interpretable baseline for regression
- No hyperparameters to tune

Why this model?

With only 35 samples, a simple model is less likely to overfit. Provides a strong baseline to compare against more complex approaches.

Method 2 — Logistic Regression

- Classifies yield as High (1) or Low (0):

$$\hat{p} = \sigma(\mathbf{w}^\top \mathbf{x} + b)$$

- Decision threshold: $\hat{p} > 0.5 \Rightarrow$ High
- Implemented with sklearn `LogisticRegression`
- Baseline for the classification task

Why this model?

Natural extension of linear regression to binary classification. Directly comparable to the neural network on the same task.

Method 3 — Neural Network (PyTorch)

Architecture: $3 \rightarrow 8 \rightarrow 1$

- Input: 3 features
- Hidden: 8 neurons, ReLU activation
- Output: 1 neuron
  - Regression: no activation, MSE loss
  - Classification: Sigmoid, BCE loss
- Optimiser: Adam (lr = 0.1)
- Epochs: 500, seed = 42

Why this model?

Can learn non-linear patterns. Kept small (3→8→1) to limit overfitting on 35 samples. Directly comparable to linear baselines.

Results — Regression (Chronological Split)

Table: Test = 2018–2024

Model	RMSE	R^2
Predict train mean	1188	−96.86
Linear Regression	**250.5**	−3.35
Neural Network	452.9	−13.22

- Linear Regression beats the naive baseline by ∼5×
- But both models still have negative R^2

Why negative R^2?

Test years (2018–2024) yield 3335–3644 kg/ha; train max is 3297. Test points sit **outside** the training range — models extrapolate. This is distribution shift, not a model-capacity failure.

Results — Regression (5-fold Cross-Validation)

Table: Mean $\pm$ std over 5 random folds

Model	**RMSE**	**R^2**
Linear Reg.	$\mathbf{164 \pm 30}$	$\mathbf{0.92 \pm 0.05}$
Neural Net	214 ± 92	0.87 ± 0.08

- When test points are interleaved with train: both models work very well
- Linear Regression edges out NN

Take-away

Same data, two protocols, opposite stories. The chronological gap was about *extrapolation*, not the models. Under fair CV, $R^2 \approx 0.92$.

Results — Classification

Table: Chronological split (test = 7/7 positive)

Model	P	R	F1
Logistic Reg.	1.00	1.00	1.00
Neural Net	1.00	1.00	1.00

Table: 5-fold stratified CV

Model	F1 (mean ± std)
Logistic Reg.	**0.933 ± 0.133**
Neural Net	0.905 ± 0.131

Why the chronological perfect score is meaningless

All 7 test years are above-median (yield grew steadily). "Always predict High" = F1 1.0. Models can't be distinguished on this split.

What CV shows

Logistic Regression slightly ahead, but within one std — difference is not significant.

Results — Comparison Plot

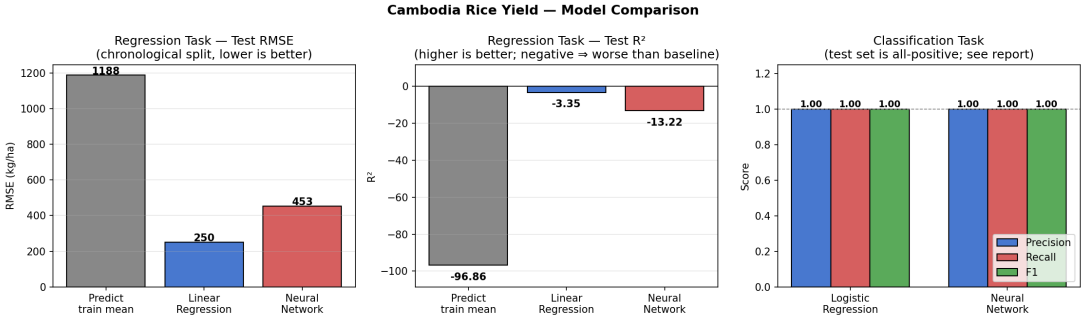


Figure: Chronological-split metrics. Left: regression RMSE; centre: regression R^2 (negative bars descend below 0); right: classification metrics (degenerate test set, see report).

Results — Cross-Validation

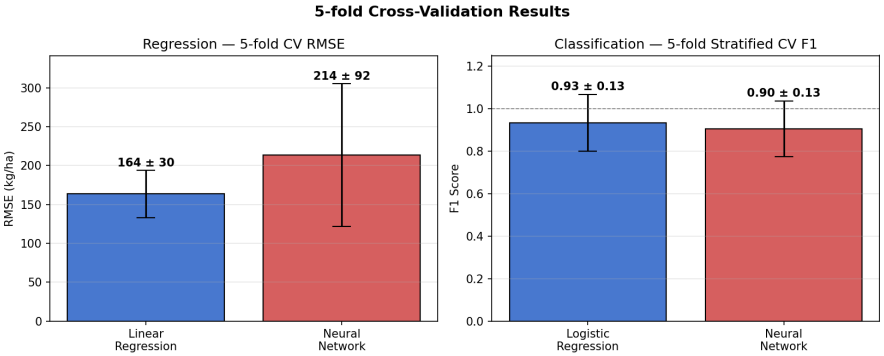
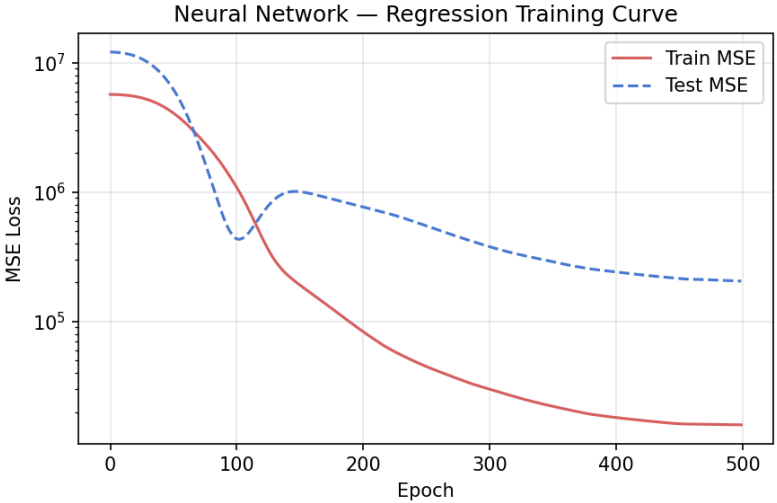


Figure: 5-fold CV gives the meaningful comparison: LR slightly ahead on both tasks; NN has higher variance.

Neural Network — Training Curve



Conclusion

Key findings:

- Under cross-validation: Linear Regression $R^2 = 0.92$, beats NN
- Chronological negative R^2 = distribution shift, not capacity
- Classification on chronological split is degenerate; CV gives the real comparison (LR $\approx$ NN)
- Linear Reg. coefficients: *area harvested* dominates (mechanisation effect)

Take-away

On a 35-row dataset, the simpler model wins under fair evaluation, and how you split matters more than which model you pick.

Future work:

- Province-level data $\Rightarrow$ $\sim$25$\times$ more rows
- Add fertiliser, irrigation, flood data
- LSTM once data volume supports it